Left-lateralization of the superior temporal gyrus during speech processing in sleeping infants predicts language skills in kindergarten: a task-based fMRI study.

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Abstract

Comparative studies have hypothesized evolutionary advantages of hemisphere asymmetries that brain lateralization facilitates cognitive skills. Previous literature has shown that speech and language are primarily left-hemisphere dominance in most individuals. However, the evidence for associations between functional lateralization and language skills has been mixed. Using functional magnetic neuroimaging (fMRI), prior studies have observed that infants already exhibit left-lateralized brain activation in the superior temporal gyrus (STG) for speech sentences, even during sleep. However, little is known about whether this earliest left lateralization is associated with concurrent and long-term language outcomes. To address the literature gap, the current study involved 24 infants who underwent fMRI during natural sleep while listening to forward- versus backward-speech when they were 3-12 months old and were subsequently assessed on language skills in preschool/kindergarten when they were 55-74 months. We observed that the left-lateralization of functional activation in STG for forward versus backward speech at infancy was associated with subsequent but not concurrent language skills. Furthermore, exploratory analyses showed that neither behavioral measures of language skills nor the left lateralization of structural brain measurements in STG at infancy were related to subsequent language skills. Overall, our findings suggest that the left lateralization of STG for speech language processing may serve as a scaffold for later language acquisition, supporting the advantageous hypothesis of hemisphere asymmetries. In addition, our finding provides preliminary evidence for a potential early neural marker for identifying children with language difficulties. However, given the small sample size, more infant task-based fMRI studies are needed to replicate our findings.

Key words: Infant, task-fMRI, speech processing, longitudinal, language development

1. Introduction

Comparative studies have suggested evolutionary advantages of hemisphere asymmetries. One assumption is that lateralized brains can avoid functional duplication and thus increase cognitive capacity (Güntürkün, et al., 2020; Rogers, 2021). Previous research has shown that speech and language-related functions are left-hemisphere dominant for most adults (Knecht et al., 2000; Carey and Johnstone, 2014) and typically developing children by 4 years old (e.g., Wood et al., 2004; Szaflarski et al., 2006; Holland et al., 2007; Olulade et al., 2020). However, there are still open questions about the emergence of speech and language lateralization and whether individual differences in brain lateralization or its developmental trajectory are related to current or future individual differences in language skills. Research has shown that the first two years of life is characterized by rapid and dynamic structural and functional brain development, which is increasingly recognized to serve as a foundation for long-term development (Gilmore, et al., 2018). Consistently, previous studies have observed that language-related white matter tracts, resting-state functional connectivity, or electrophysiological responses at infancy were predictive of subsequent language skills (e.g., Zuk et al., 2021; Yu, et al., 2021; Lohvansuu et al., 2021). However, whether speech and language lateralization at infancy is prospectively related to long-term language skills remains largely unknown. Addressing this question will allow us to examine the advantageous hypothesis of hemisphere asymmetries from a speech language domain. In addition, it might provide preliminary evidence for early neural markers for later language development if robust predictors are observed.

With the advancement of various neuroimaging techniques including functional magnetic resonance imaging (fMRI), and near-infrared spectroscopy (NIRS), previous studies have investigated infant brains to trace the earliest neural activity during speech language processing. The literature seems to suggest a clear left lateralization that emerges early in infancy when processing speech sentences compared to processing non-speech sounds (e.g., Dehaene-Lambertz et al., 2010; Shultz et al., 2014; Minagawa-Kawai et al., 2011). For example, using fMRI, Dehaene-Lambertz et al. (2010) showed that speech sentences elicited greater activation in the left temporal region compared to music, whereas music induced greater activation in the right planum temporale than speech in 2-month-old infants. However, when a task involves speech sentences that are manipulated (reversed or flattened) or from another unfamiliar language, observations of hemispheric dominance in infants have been mixed (e.g., Dehaene-Lambertz et al., 2002; Pena et al., 2003; Vannasing et al., 2016; May et al., 2011; 2018; Perani et al., 2011; Zhang et al., 2022). For example, some studies continued to observe left-lateralized brain activity in the temporal areas for forward speech sentences of their native languages using fMRI (e.g., Dehaene-Lambertz et al., 2002) and NIRS (Pena et al., 2003; Vannasing et al., 2016) when infants were also presented with backward speech sentences (e.g.,

Dehaene-Lambertz et al., 2002; Pena et al., 2003) or speech sentences from a foreign language (e.g., Vannasing et al., 2016). Others, however, observed no hemisphere dominance (e.g., May et al., 2011), right hemisphere dominance (e.g., Perani et al., 2011), or mixed findings of hemispheric dominance (e.g., May, et al., 2017; Zhang et al., 2022). Together, these studies suggest that speech language is likely processed with left hemisphere dominance as early as infancy, but infants' brain lateralization for speech language processing may vary when fine-tuned linguistic discrimination is needed.

Although speech language processing is primarily left-hemisphere dominance in many people, and maybe in infants, it is unclear whether the extent of this hemispheric dominance is linked with typical and/or atypical language skills. Using fMRI or functional transcranial Doppler ultrasound (fTCD), studies on older children beyond 5 years old have mixed findings in the relationship between brain lateralization and language skills. In studies on typically developing children using various language tasks, some observed a positive relationship between left lateralization of language processing and language skills (e.g., Everts et al., 2009; Greon et al., 2012; Younger et al., 2019), while others showed a negative correlation (e.g., Bartha-Doering et al., 2018), or mixed (i.e., positive, negative or no) correlations depending on tasks (e.g., Lidzba et al., 2011), or brain regions (e.g., Berl et al., 2014). In studies on children with developmental language disorder (DLD), some early studies supported a decreased left-lateralization in children with DLD (e.g., De Guibert et al., 2011; Badcock et al., 2012). However, recent studies with larger sample sizes showed no difference between groups in hemispheric dominance and no brain-behavioral correlations (e.g., Vansteensel et al., 2021; Krishnan et al., 2021; Wilson & Bishop, 2018). In contrast to a myriad of studies on older children, although the first five years is an important period for language development, studies on the relationship between functional lateralization for speech language processing and language skills in children younger than 5 years old are few. Among the limited studies on younger children (e.g., 4-year-olds in Bishop et al., 2014; 1- to 5-year-olds in Kohler et al., 2015; 6-month-olds in Cantiani et al., 2019), the studies by Bishop et al. (2014) and Cantiani et al. (2019) consistently showed atypical lateralization for speech or auditory processing in children with DLD (or a familial risk of DLD) as compared to children without DLD (or a familial risk of DLD), suggesting an association between brain lateralization and atypical language skills at the population level. However, the studies by Bishop et al. (2014) and Kohler et al. (2015), which examined the correlations between individual differences of left lateralization and individual differences of language skills, did not observe a significant effect. Overall, previous studies on younger children primarily used fTCD and EEG (e.g., fTCD in Bishop et al., 2014 and Kohler et al., 2015; EEG in Cantiani et al., 2019), which have low spatial resolution. If the lateralization in the brain is fine-grained and focally located, studies using these techniques may have low sensitivity in capturing individual

differences in brain lateralization for language processing. In addition, most previous studies in younger children (e.g., Bishop et al., 2014; Kohler et al., 2015) and all previous studies in older children (e.g., Everts et al., 2009; Berl et al., 2014; Greon et al., 2012; Younger et al., 2019; Lidzba et al., 2011; Bartha-Doering et al., 2018; De Guibert et al., 2011; Badcock et al., 2012; Vansteensel et al., 2021; Krishnan et al., 2021; Wilson & Bishop, 2018; Bishop et al., 2014) used active language tasks. The use of active language tasks may elicit children's strategic processes to achieve high language performance but confound their brain left lateralization by engaging multiple areas, and thus reduce the likelihood of observing brain-behavioral correlations. Besides, all previous studies (Everts et al., 2009; Berl et al., 2010; Greon et al., 2012; Younger et al., 2019; Lidzba et al., 2011; Bartha-Doering et al., 2018; De Guibert et al., 2011; Badcock et al., 2012; Vansteensel et al., 2021; Krishnan et al., 2021; Wilson & Bishop, 2018; Bishop et al., 2014; Kohler et al., 2015; Cantiani et al., 2019) only examined the concurrent relationship between language lateralization and language skills/risk status, little is known about whether language lateralization serves as a foundation for subsequent language acquisition which requires a longitudinal design to address.

To better address whether left lateralization of speech language processing contributes to language skills in early childhood, it would be valuable to examine infant brain responses during passive language tasks, which allows us to examine the earliest language lateralization with little confounds of strategic processes. It would also be important to use fMRI, which offers better spatial resolution to capture variances in language lateralization if it is more fine-grained and focal. In addition, longitudinal designs are needed to clarify the directionality of the relationship. To the best of our knowledge, despite a few fMRI studies suggesting an early emergence of left lateralization (Dehaene-Lambertz et al., 2010; Shultz et al., 2014; Dehaene-Lambertz et al., 2002), no fMRI studies have examined whether the left lateralization of language processing in infants is related to their concurrent and/or future language skills.

To fill the literature gap, the current study used a longitudinal design and investigated the relationship between infants' brain lateralization for speech language processing using fMRI during sleep and their subsequent language skills at preschool/kindergarten. A canonical speech language processing fMRI task that included both forward and backward speech blocks was employed to evaluate neural activity for speech language processing at infancy. We focused on the lateralization of activation in the superior temporal gyrus (STG) because this area is thought to be associated with speech perception and phonological processing (Hickok & Poeppel, 2000; Leonard & Chang, 2014) and is consistently found to be engaged in speech language processing tasks in infants during natural sleep (Dehaene-Lambertz et al., 2010; Shultz et al., 2014; Dehaene-Lambertz et al., 2002). Language skill was assessed at preschool/kindergarten because major milestones of language such as phonological, semantic, and

syntactic skills have already developed by this age and become stabilized (Gervain, 2021). We also measured children's language skill at infancy to control for autoregressive effects. We hypothesized that, if brain lateralization has evolutionary advantages (Güntürkün, et al., 2020; Rogers, 2021), the extent of left lateralization for speech language processing in STG at infancy would predict later language skills at preschool/kindergarten, even after controlling for the autoregressive effect of language skills at infancy.

2. **Method**

2.1. Participant

Twenty-four children (13 females, 11 males) were included in the current study for final analyses. All children underwent a speech task fMRI at infancy during natural sleep when they were 3 to 12 months old (mean age = 7.6, SD = 3.4) and their language skills in preschool or kindergarten when they were 55 to 74 months old (mean age = 61.0, SD = 4.1). All the following procedures were approved by the institutional review boards at Boston Children's Hospital and Harvard University. Before participation, informed written consent was obtained from a parent or legal guardian of each child.

2.2. Procedure

Language skill was measured at infancy using the language composite (i.e., expressive + receptive language) score in the Preschool Language Scale Fifth Edition (PLS-5, Zimmerman, Steiner, & Pond, 2011) and assessed again at preschool or kindergarten age using the oral expressive score in the Woodcock Johnson IV Tests of Oral Language. We used the oral expressive score in the Woodcock Johnson IV Tests of Oral language (Schrank, Mather, & McGrew, 2014) out of the 3 composite language skills (i.e., oral expressive, listening comprehension, and phonetic coding) because it has the largest sample size, which increases the power of conducting correlational analyses.

A block-design speech task fMRI was carried out in children at infancy during their natural sleeping. In this speech task, a female-voice recorded excerpt of Dr. Seuss's "One Fish, Two Fish" was played through headphones while infants were asleep. The audios were presented in alternating forward and backward speech blocks that each lasted about 18 seconds. Rest periods of 18 seconds were inserted randomly. In the whole task, there were 6 blocks of forward speech, 6 blocks of backward speech and 4 blocks of rest. An extra 7.2s fixation screen was included at the beginning of the task to achieve contrast stability. Infants first completed the standardized behavioral battery and then underwent MRI. Behavioral testing was done first to tire the infants out and increase the likelihood that they would fall asleep prior to scanning. When they fell asleep, they were then put in the scanner (see details of infant scan in Raschle et al., 2012).

This study was part of a large longitudinal project from the National Institute of Health (R01 HD065762). There were originally 59 children who participated in the fMRI speech task at infancy. However, only 25 of them were obtained with composite language scores from the Woodcock Johnson IV Tests of Oral Language. Specifically, we had 25 children with oral expressive scores, which tested on picture vocabulary and sentence repetition skills, 11 children with listening comprehension scores, which measured oral comprehension and listening directions, and 20 children with phonetic coding scores, which assessed segmentation and sound blending. Of the 25 children who had both brain measurement at infancy and language measurement at preschool or kindergarten, one child whose standard score of the oral expressive score was below 2.56 standard deviations of the mean was treated as an outlier and excluded from our final analyses. Therefore, 24 participants were finally included in the current study.

2.3. Scanning parameters

All brain images were acquired between 2016 and 2019 utilizing a Siemens scanner at Boston Children's Hospital with a 32-channel head coil. T1-weighted image was acquired using a sequence of Magnetization-Prepared Rapid Acquisition Gradient Echo (MPRAGE) with the following parameters: TR = 2520 ms, TE = 1.74 ms, flip angle =7, field of view = 192, slice thickness = 1 mm, voxel size = 1 mm x 1 mm x 1 mm, matrix size = 192 x 192, number of slices = 144. Functional images were acquired using a sequence of echo planar imaging (EPI) with the following parameters: TR = 720 ms, TE = 30 ms, flip angle = 60, multiband acceleration factor = 3, field of view = 192, slice thickness = 4 mm, voxel size = 3 mm x 3 mm x 4.8 mm, matrix size = 64 mm x 64 mm, number of slices = 33, acquisition sequence = interleaved from foot-to-head.

2.4. Data analyses

sMRI preprocessing.

Infant FreeSurfer image analysis suite (Zollei et al., 2020) was be used to extract the brain from the skull, correct for intensity inhomogeneity, and segment T1 images into different tissue classes (gray matter, white matter, cerebrospinal fluid) and brain regions. To improve tissue classification accuracy, T1 images were submitted in parallel to iBEATv2.0 Docker (Wang et al., 2018; Li et al., 2014; 2015; 2018). Resulting segmentations from each software package were then combined and resubmitted to a modified FreeSurfer "recon" pipeline to generate white matter and pial surfaces, and then regional estimates of gray matter volume.

fMRI preprocessing.

Before conducting any fMRI preprocessing, the first 10 functional volumes were discarded. All functional images were then realigned to the mean functional image using Statistical Parametric Mapping

(SPM12). Co-registration of functional images to their corresponding skull-striped T1-weighted image was estimated using the Advanced Normalization Tools (ANTs) with a rigid transformation, but functional images were not resampled at this step. After that, the skull-stripped T1-weighted image was warped to the standard space of the 1-year-old pediatric anatomical template developed as part of the UNC 0-1-2 Infant Atlases (Shi et al., 2011) using ANTs with both affine and syn deformable transformations. We then applied the transformations (rigid co-registration + affine and syn deformable normalization) to the functional images such that they were normalized to the 1-year-old pediatric template using one interpolation. Finally, smoothing was applied to all normalized functional images using a 4-mm Gaussian kernel. Artifact Detection Tools (ART) was used to identify outlier volumes with greater than 1.5-mm scan-to-scan displacement or 4% global signal change from the mean. All participants had no more than 20% outlier volumes.

First-level modeling and lateralization calculation

A general linear model (GLM) was generated using SPM12 by entering the onsets and durations for the forward speech, backward speech, and rest conditions as covariates of interest. Motion parameters, artifact regressors, and global mean signal generated by ART were entered into the GLM as covariates of no interest. The hemodynamic function (HRF) in SPM12 was replaced with an infant HRF which had protracted and attenuated positive peak with an increased undershoot to peak ratio compared with adults (Arichi, et al., 2012). The contrast of forward > backward speech was generated to evaluate speech language processing in the brain. To calculate the left lateralization of brain activation in the superior temporal gyrus (STG), left and right STG masks were first generated using the UNC 1-year-old Neonate Automated Anatomical Labeling atlas (Shi et al., 2011). Brain activation for the contrast of forward > backward speech was then extracted from the most activated 100 voxels in the left and right STG respectively. The left lateralization index of brain activation in STG was calculated using (left – right)/(left + right) for each participant. To ensure engagement of bilateral STG during natural sleep, we conducted a one-sample t-test for contrasts of forward speech > rest, backward speech > rest, and forward > backward speech within the bilateral STG mask using a lenient threshold of voxel-wise $p < 0.05$, cluster size > 50.

Correlational analyses

Pearson correlation was first used to evaluate if the left lateralization of brain activation in STG for speech language processing during natural sleep at infancy was associated with later expressive language scores. Age was included in the correlational table to examine if brain lateralization indices or standard language scores were impacted by age. To examine the directionality of the influence, a hierarchical

regression analysis, controlling the autoregressive effect of language composite scores at infancy, was further conducted.

Exploratory analyses

Although there was little support for a link between gray matter asymmetry in STG and language skills in previous literature (see review by Mayes, Reilly, & Morgan, 2015), no previous studies have examined the longitudinal relationship. Given that T1-weighted anatomical images were collected for all participants, we conducted parallel analyses to examine if the left lateralization of brain structure (i.e., surface area, cortical thickness, gray matter volume) of STG at infancy was related to later expressive language scores using Pearson correlation.

3. Results

3.1. Description of Age, Language skills, and Brain Activation in STG

The mean, standard deviation, and range for children's age in months and language skills at both infancy and preschool/kindergarten time points are displayed in **Table** 1. In terms of brain activation, we observed that both the left and right STG were activated for the contrasts of forward speech > rest, backward speech > rest, and forward > backward speech, suggesting that bilateral STG were engaged during the speech fMRI task even when infants were asleep (**Table** 2 and **Figure** 1). The left lateralization of brain activation in STG for forward > backward speech had a lateralization index mean value of -0.08, standard deviation of 0.21 and range from -0.47 to 0.24. A positive value indicates left lateralization, whereas a negative value means right lateralization.

Table 1. Description of children's age in months and language skills at both infancy and preschool/kindergarten time points.

Variables	Mean	Standard Deviation	Range
Age in months at infancy	7.6	3.4	2.5 – 12.9
Age in months in preschool/kindergarten	61.0	4.1	55.0 – 69.1
Language composite score at infancy	95.2	9.3	73 – 112
Expressive language score in preschool/kindergarten	114.2	10.6	96 – 133

Note: Language composite score at infancy was the composite score of both expressive and receptive languages from PLS-5; Expressive language score in preschool/kindergarten was the standard score from Woodcock Johnson IV Tests of Oral language.

Table 2. Brain activation in the left and right STG for contrasts of forward speech > rest, backward speech > rest, and forward > backward speech.

Contrasts	Left Superior Temporal Gyrus		Right Superior Temporal Gyrus	
	T value of peak	Cluster size (Num. of voxels)	T value of peak	Cluster size (Num. of voxels)
Forward speech > Rest	3.76	361	3.21	340
Backward speech > Rest	4.76	273	2.58	146
Forward > Backward speech	2.58	76	4.48	309

Note: the MNI coordinates were not reported due to the use of infant standard space.

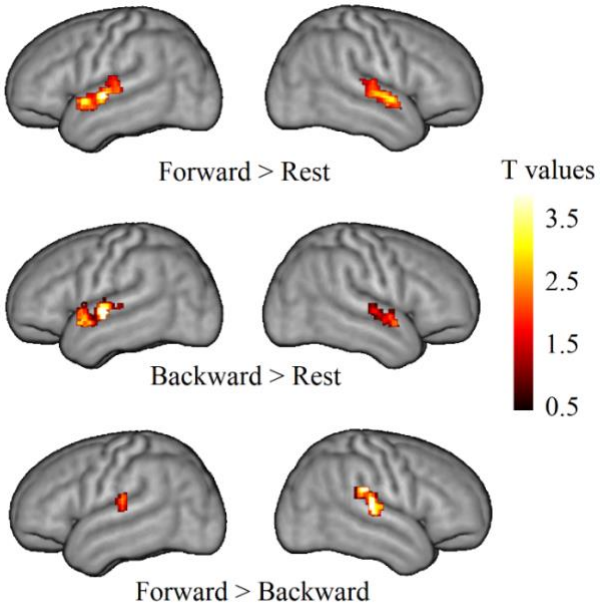

Figure 1. Brain activation within the left and right STG masks for contrasts of forward speech > rest, backward speech > rest, and forward > backward speech at a threshold of voxel-wise $p < 0.05$, cluster size > 50, uncorrected.

3.2. Brain Behavioral Correlations

Pearson correlations between brain and language variables at infancy and preschool/kindergarten were computed (**Table** 3). We also added age at both time points into the correlation table to examine if brain lateralization and the standardized score of language skills were influenced by age. We observed no correlations between age and brain lateralization or the standardized scores of language skills, suggesting age was not a factor that needed to be controlled for in the following regression analyses. As for our analyses of interest, we observed that left lateralization of brain activation in STG during infancy was

prospectively correlated with expressive language at preschool/kindergarten age ($p < .001$). Results showed that the higher the left lateralization of brain activation in STG in infancy, the better children's expressive language skill at preschool/kindergarten age (**Figure 2**). However, we did not observe a significant correlation between language composite score at infancy and the expressive language score at preschool/kindergarten age ($p = .075$). In addition, the left lateralization of brain activation in STG at infancy was not correlated with language composite score in infancy ($p = .126$). Hierarchical regression analysis showed that after controlling the effect of language composite score in infancy, the left lateralization of brain activation in STG at infancy still significantly predicted expressive language score at preschool/kindergarten age, explaining an additional 33.4% variances (**Table 4**).

Table 3. Pearson correlations among different variables.

Variables	2	3	4	5
1.Age in months at infancy	.181 (.397)	.146 (.496)	.170 (.427)	-.129 (.547)
2.Age in months in preschool/kindergarten	1	-.020 (.925)	.068 (.753)	-.107 (.620)
3.Left lateralization of brain activation in STG		1	.321 (.126)	**.666 (<.001)**
4.Language composite score at infancy			1	.370 (.075)
5.Expressive language score in preschool/kindergarten				1

Note: values outside the parenthesis are r values whereas values inside the parenthesis are *p* values. Bolded values indicate significant correlations. *p* values displayed were not corrected for multiple comparisons.

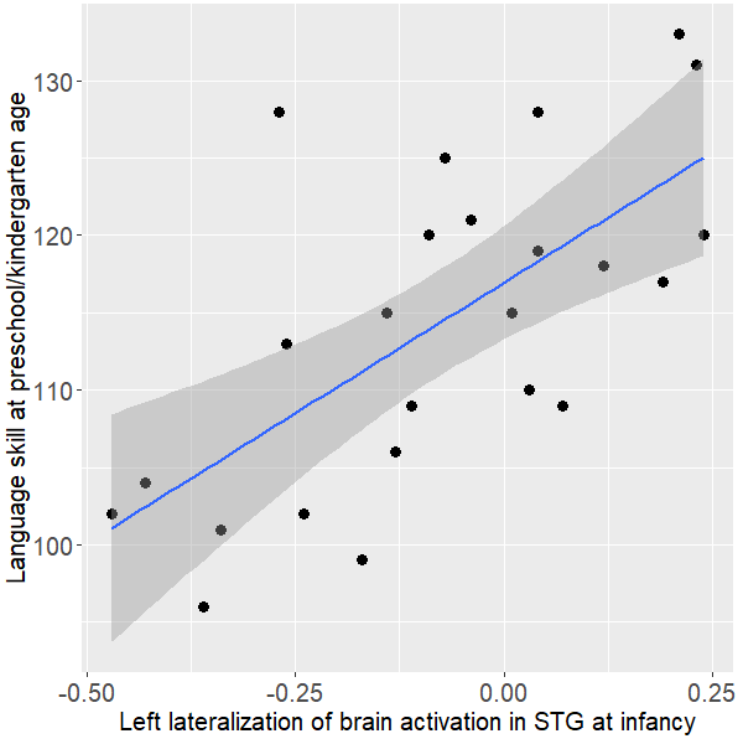

Figure 2. Scatterplot of the relationship between the left lateralization of brain activation in STG at infancy and language skill at preschool/kindergarten age. Language skill at preschool/kindergarten age was the standardized score of expressive language. Left lateralization of brain activation in STG at infancy was a ratio calculated by brain activation of (left - right)/(left + right). A positive value indicates left lateralization, whereas a negative value means right lateralization.

Table 4. Hierarchical regression of the relationship between the left lateralization of brain activation in STG at infancy and expressive language score at preschool/kindergarten age, after controlling language composite score at infancy.

Models	Predictors	Dependent variable			
		Expressive language score in preschool/kindergarten			
		Beta	*p*	R^2	R^2 change
1	Language composite score at infancy	.370	.075	.137	-
2	Language composite score at infancy	.174	.311		
	Left lateralization of brain activation in STG	**.610**	**.002**	.470	**.334**

Exploratory analyses did not show any significant longitudinal correlations between the left lateralization of brain structure (i.e., surface area, cortical thickness, gray matter volume) in STG at infancy

and expressive language score at preschool/kindergarten age (*p*s >= .556). There were no significant concurrent correlations between structural lateralization in STG and language composite score at infancy either (*p*s >= .219).

4. Discussion

The current study examined the longitudinal relationship between the neural correlates of language lateralization at infancy and their subsequent language skill at preschool/kindergarten using fMRI. We observed that early left-lateralization index in the superior temporal gyrus (STG) for forward versus backward speech at infancy significantly predicted subsequent expressive language score at preschool/kindergarten, even after controlling for initial language composite score at the infancy timepoint. In contrast, neither language skill nor anatomical lateralization index in STG measured at infancy were correlated with expressive language score at preschool/kindergarten. These findings support the advantageous role of early language lateralization in subsequent language acquisition, providing preliminary evidence of a potential neural index for early identification of children who may have subsequent language difficulties.

The key finding of the current study is that functional lateralization of STG at infancy was significantly associated with expressive language skills 4-5 years later. Compared with previous literature on children younger than 5 years old which had weak or no evidence for the association between left lateralization of speech language processing and language skills (Bishop et al., 2014; Kohler et al., 2015; Cantiani et al., 2019), the current study had several advantages that may have increased the sensitivity of capturing the relationship between brain lateralization and language skills. Firstly, we examined infants' brain lateralization using a passive listening task during natural sleep, which allowed us to examine the earliest functional lateralization during speech language processing with minimum confounds from strategic processes induced by active language tasks. Second, unlike previous studies using EEG or fTCD, which has low spatial resolution, the use of fMRI allowed the current study to more precisely localize the region of interest to calculate brain lateralization. Specifically, we used brain activity from the top activated voxels (~ 5%) within STG for each individual, instead of using the whole STG or group activated STG, to calculate the lateralization index. This individualized top-voxel approach has been shown to be more sensitive in finding brain-behavioral correlations across language, reading, and math domains (e.g., Bitan, et al., 2009; Younger et al., 2019; Gullick, & Booth, 2014; Suarez-Pellicioni, et al., 2020; Yamasaki, et al., 2021; Wang et al., 2020; 2021) and has successfully been employed in several recent infant fMRI studies in detecting

subtle functional specialization (e.g., Deen et al., 2017; Kosakowski et al., 2023; Yates, Ellis, & Turk-Browne, 2023).

The third advantage of the current study is that it examined the longitudinal relationship between brain lateralization for speech language processing and language skills. The fact that we observed a significant scaffolding effect of early brain lateralization at infancy for later expressive language skills provides the first longitudinal evidence for a potential advantageous role of brain lateralization in language development. In contrast to finding a significant longitudinal relationship, we observed no concurrent correlations between brain lateralization and language skill at infancy, consistent with the lack of correlations in observed in previous studies on children younger than 5 years old (Bishop et al., 2014; Kohler et al., 2015) and other recent studies on older children with larger sample sizes (e.g., Vansteensel et al., 2021; Krishnan et al., 2021; Wilson & Bishop, 2018). One possibility for the lack of a concurrent correlation observed in the current study is that infants acquire the ability to produce their first words at approximately 12 months of age (Kuhl, 2004), making it very challenging to examine early language skills. Thus, behavioral measures of language composite skills at infancy obtained primarily by experimenters observing infants' behavioral performance in labs may not be as sensitive as brain measures in capturing individual differences in their ability for speech language processing. Consistent with this speculation, we observed that only brain lateralization but not behavioral measures of language composite score at infancy significantly predicted subsequent expressive language skills measured at preschool/kindergarten. Another reason for the lack of concurrent correlations observed in the current study on infants and in previous studies on older children could be due to that some children had a complementary use of right hemisphere (regardless of consciousness), which reduced their brain left lateralization but facilitated their concurrent speech language processing skills. In this case, we hypothesize that a more left lateralized brain function for speech language processing at an earlier developmental time point might not be sensitive in reflecting concurrent language skills due to a potential complementary mechanism of right hemisphere engagement, but a stronger left lateralization likely serves as a better scaffold for language acquisition in subsequent years. However, more longitudinal studies with different age ranges and much larger sample sizes are needed to examine this hypothesis.

In the current study, we observed that although brain activity for both forward and backward speech as compared to rest is left-lateralized in STG, there seems to be a right-lateralized activation in STG for forward versus backward speech at the group level. The observed left-lateralized activation in STG for both forward and backward speech compared to rest is consistent with the majority of previous research on infants (Dehaene-Lambertz et al., 2010; Shultz et al., 2014; Minagawa-Kawai et al., 2011; Dehaene-

Lambertz et al., 2002; Pena et al., 2003; Vannasing et al., 2016), which suggests that processing speech sentences (i.e., both forward and backward speech) are likely already left-lateralized in infants. Because brain activity may be confounded by basic auditory processing when simply using the contrast of forward (or backward) sentences versus rest, the current study employed the contrast of forward versus backward speech to evaluate functional lateralization of STG for speech language processing. The forward and backward speech shared basic auditory information but only the former one was intelligible, containing phonological, semantic, and syntactic information. However, we observed a slightly right-lateralized activation in STG when using the contrast of froward speech versus backward speech. This observation is not surprising because in previous literature, when the task involved speech sentences that were manipulated (reversed or flattened) or from another unfamiliar language, the findings of brain lateralization for speech processing were also mixed (Dehaene-Lambertz et al., 2002; Pena et al., 2003; Vannasing et al., 2016; May et al., 2011; 2017; Perani et al., 2011; Zhang et al., 2022). Previous literature has shown that phonological, lexico-semantic processing tends to be left hemisphere dominant, whereas prosody is right hemisphere dominant in both adults and infants (e.g., Price, 2012; Arimitsu et al., 2011). We suspect that in sleeping infants, the left-hemispheric sensitivity to fine-grained linguistic information constrained in their native language is not fully developed. When presented with clear speech versus non-speech stimuli (or rest), infants can consistently engage left-lateralized STG to process the acoustic and linguistic information. However, when presented with manipulated speech stimuli that require fine-grained linguistic processing, infants whose linguistic representation in the left STG is coarse may need to rely on prosody represented in the right STG to help increase the contrast between their native versus manipulated (reverse or flattened) speech. In the current study, despite infants showing an overall right-lateralized activation, which may indicate reliance on prosody, individuals who showed stronger left-lateralization at infancy, likely indicating those who had more fine-grained linguistic representation of their native language, had better subsequent language skills at preschool/kindergarten.

Although functional lateralization of STG was observed as an early predictor for subsequent language skills, parallel exploratory analysis revealed that anatomical lateralization of STG was not. Some previous studies examined the correlation between structural measurements and language skills. However, they either used different brain measurements (e.g., Aeby et al., 2013) or observed an effect in other brain regions (e.g., Qi et al., 2019). Aeby et al. (2013) showed that the mean diffusivity (MD) of the left STG using diffusion tensor imaging (DTI) in 41 pre-term infants at term equivalent age was associated with children's language skills at 2 years old. Qi et al. (2019) observed that the left lateralization of cortical thickness in the inferior frontal lobe in 76 5-6-year-old children was correlated with language skills at 7 years old. In

379 studies with atypical population, although several previous studies occasionally demonstrated atypical
380 (more or less) grey matter volume in STG from children with DLD as compared to typically developing
381 children (Asaridou & Watkins, 2022), among the studies that examined structural asymmetry, the majority
382 reported that children with DLD do demonstrate typical left hemisphere structural asymmetry within
383 temporal language regions including the pSTG, similar to that seen in typically developing children (Mayes,
384 Reilly, & Morgan, 2015). Although structure is typically thought to underlie function, our study observed
385 that structural lateralization of STG using surface area, volume, and cortical thickness at infancy was not
386 associated with language skills either concurrently or longitudinally. This additional finding seems to
387 suggest that only functional but not structural lateralization of STG is sensitive in predicting subsequent
388 language skills. However, it could also be due to small sample size, insensitivity of the structural
389 measurements chosen, and/or brain regions selected that more future studies are needed to address.

390 There are several limitations of the current study. First, due to the extreme difficulty in collecting infant
391 fMRI and longitudinal dataset that lasted approximately 5 years, the current study ended up with a
392 relatively small sample size. More future studies with larger sample sizes are needed to examine if the
393 findings of our current study are replicable. Second, although most people develop left-lateralized
394 language processing in the brain, studies have found that people with right or no functional hemisphere
395 dominance also show typical language skills (Knecht et al., 2000; Carey and Johnstone, 2014). This current
396 study was conducted under an assumption that the participants included in our study followed a typical
397 left hemisphere dominance trajectory for language processing and thus we observed as expected that the
398 more left-lateralized at infancy for language processing, the better their subsequent language skill was.
399 However, we do not know whether the rare cases of opposite or no hemisphere dominance were included
400 in our study. Third, all the infants were sleeping during the fMRI scan. Previous research suggests that
401 sleeping as compared to awake status leads to reduced brain activity for linguistic information, especially
402 the higher-order frontal regions (e.g., Wilf et al., 2016). The use of the lateralization index, which scaled
403 brain activity by using (left – right)/(left + right), likely reduced the impact of sleeping stage on the
404 individual differences of brain engagement/activation. However, we cannot make sure that the sleeping
405 status had no impact on brain lateralization. Future studies scanning awake infant with fMRI are needed
406 to confirm the findings of our current study.

407 In summary, the current study used fMRI to examine the longitudinal relationship between language
408 lateralization of STG in sleeping infants and their subsequent expressive language skill at
409 preschool/kindergarten, an age by which children have gained the major aspects of language such as
410 phonological, semantic, and syntactic skills. We observed that functional lateralization in STG for speech

sentence processing at infancy predicts later expressive language skill, supporting the potential of a scaffolding role of earlier language lateralization on subsequent language development. Parallel exploratory analyses revealed that neither language skill measured behaviorally nor with an anatomical lateralization index in STG at infancy predicted later language skill. These findings provide preliminary evidence that the functional lateralization index of STG for speech language processing in infants might be a more sensitive early neural marker than the behavioral or parallel anatomical measurements in terms of identifying children who may later struggle with language acquisition. However, before reaching a solid conclusion, more fMRI studies with larger sample sizes, awake infants, and longitudinal design are needed to examine the replicability of the current study.

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Data code availability statement

The data used in this study is accessible upon request. The code used for this study is shared on GitHub (https://github.com/wangjinvandy/infant_fMRI_lateralization_predicts_language_skil_at_preschool-kindergarten).

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